

APOORV AGRAWAL

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EDUCATION

Bachelor of Science in Computer Engineering

Texas A&M University

May 2029 (expected)

College Station, TX

TECHNICAL SKILLS

Languages: C++, C, Python, Java

Tools: KiCad, Altium, STM32CubeIDE, SEGGER Ozone, oscilloscope, logic analyzer, multimeter, Git, Fusion360

Embedded Systems: STM32, ARM Cortex-M Architecture, DMA, I2C, UART, CAN

Peripherals & Control: Sensor interfacing, motor controllers, PID/feedforward loops

EXPERIENCE

Software Developer | C++, Python, Visual Studio

August 2025 – Present

Aggie Robotics

College Station, TX

- Implementing a real-time Informed RRT* + B-Spline planner in C++ achieving sub-50 ms trajectory generation in dynamic 2D environments
- Integrating a spatial 2D Kd-Tree and heuristic ellipsoidal sampling, improving path generation speed by 4x
- Designing a modular API for easy integration with rest of the codebase and sensor-driven obstacle updates

Firmware Developer | C++, SEGGER Ozone

September 2025 – Present

Texas Aimbots

College Station, TX

- Executing a large-scale migration of a 10k-LOC codebase to a breaking library update in a team of three, resolving compatibility issues and modernizing the stack
- Interfacing with sensors and motor controllers to implement and tune feedback and feedforward control loops in C++ for real-time motion control, significantly decreasing response latency

Coding Camp Team Lead

June 2025 – July 2025

Prosper ISD

Prosper, TX

- Led 3 instructors to teach 40+ students (ages 8-12) Python and Scratch programming fundamentals
- Coordinated with supervisors to develop lesson plans and schedules for the week

Co-lead Programmer | Java, Fusion360

August 2021 – June 2025

FIRST Robotics

Plano, TX

- Placed 2nd in Texas (2023), Qualified to World Championships (2025)
- Mentored and trained 3 rookie members in Java and basic linear control systems
- Presented research on Gain Scheduling (Linear control for non-linear systems) at a regional workshop
- Integrated and installed sensors (quadrature encoders, gyroscopes, color sensors, ultrasonic sensors)

PROJECTS

Dynamic Survey Tool | Altium, STM32CubeMX

September 2025 – Present

- Collaborating with a team of 5 to develop an MWD (Measure-While-Drilling) PCB for subsurface orientation sensing
- Researching and implementing filtering and estimation algorithms (Low-pass filter, Extended Kalman Filter) on an STM32 microcontroller
- Created schematics and routed high-density PCB traces to simulate multiplexing on a Parallel NOR Flash using external 8-bit latches for integration with the STM32 Flexible Memory Controller

Raspi Chatbot | Python, Debian Linux, Git

April 2025 – May 2025

- Designed and built a voice-interactive chatbot using Raspberry Pi, LCD panel, microphone, amplifier, speaker, and push-button interface
- Implemented end-to-end workflow: device activation via button press → speech input capture → AI response generation → text-to-speech output
- Developed a custom logging utility to track and verify data flow, ensuring system reliability and easier debugging
- Integrated multiple libraries including Ollama, pyttsx3, and speech_recognition

Forza Motorsport Telemetry Display | Avalonia, C#, Git

July 2023 – January 2024

- Developed a desktop dashboard app for Forza Motorsport 7 to display real-time car telemetry transmitted over UDP
- Reverse-engineered undocumented packet structures to extract relevant data